

COMPANION
to

ChatGPT Teaches TikZ

Chapter 12: Matrix-Based Diagrams (I)

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12.1 What This Companion Adds

Chapter 12 treated a TikZ matrix as a layout engine rather than merely a typeset array. This Companion develops that viewpoint by separating four decisions: matrix structure, spacing, common cell style, and exceptional cells or connections.

Design

Let the matrix determine the row-and-column geometry.
Add coordinates only for objects that genuinely lie outside
the matrix structure.

12.2 A Matrix Vocabulary

Form or option	Purpose
<code>matrix of nodes</code>	Makes every entry a TikZ node.
<code>row sep=6mm</code>	Sets spacing between rows.
<code>column sep=10mm</code>	Sets spacing between columns.
<code>nodes={...}</code>	Applies options to every entry.
<code>minimum width</code>	Sets the minimum cell width.
<code>minimum height</code>	Sets the minimum cell height.
<code>(m)</code>	Gives the complete matrix a name.
<code>m-2-3</code>	Refers to row 2, column 3.
<code> [options] </code>	Styles one entry differently.

The matrix library is loaded in the preamble:

```
\usetikzlibrary{matrix}
```

12.3 Reading Matrix Syntax

Within the matrix body, an ampersand separates columns and a double backslash ends a row.

```
\matrix[matrix of nodes]
{
  A & B & C \\
  D & E & F \\
};
```

The final semicolon terminates the complete matrix command.

Common Mistake

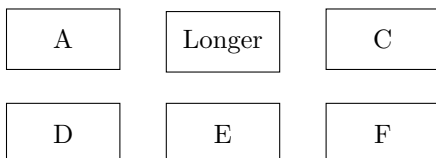
Do not omit the double backslash after the final visible row. It terminates that row; the semicolon then terminates the matrix command.

12.4 A Uniform Grid

Shared node options keep entries with different text lengths visually aligned.

```
\begin{tikzpicture}
  \matrix[
    matrix of nodes,
    row sep=5mm,
    column sep=7mm,
    nodes={
      draw,
      minimum width=16mm,
      minimum height=8mm
    }
  ]
  {
    A & Longer & C \\
    D & E       & F \\
  };
\end{tikzpicture}
```

See



A	Longer	C
D	E	F

Use `minimum width` and `minimum height` for consistent cells; use `row` and `column` separation for the space between cells.

12.5 Styling Every Entry

The `nodes` option describes the ordinary cell once.

```

nodes={
  draw,
  rounded corners,
  fill=blue!10,
  minimum width=14mm,
  minimum height=8mm
}

```

This is more maintainable than repeating four or five options in every entry.

12.5.1 Highlighting One Entry

Place a local option immediately before the entry:

```

1 & 2 & 3 \\
4 & |[fill=yellow]| 5 & 6 \\
7 & 8 & 9 \\

```

The shared options still apply; the local fill marks one exception.

See

1	2	3
4	5	6
7	8	9

Negative separation by one line width makes neighboring borders meet cleanly.

12.6 Naming and Addressing Entries

Give the matrix a name before referring to cells.

```
\matrix (m) [matrix of nodes]
{
  A & B \\
  C & D \\
};
```

TikZ automatically creates the names `m-1-1`, `m-1-2`, `m-2-1`, and `m-2-2`.

Name	Entry
<code>m-1-1</code>	First row, first column.
<code>m-1-2</code>	First row, second column.
<code>m-2-1</code>	Second row, first column.
<code>m-2-2</code>	Second row, second column.

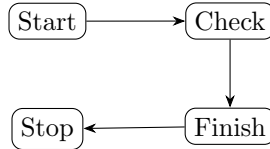
The name describes location, not visible content.

12.7 Connecting Matrix Entries

Because each entry is a node, ordinary TikZ paths can connect them.

```
\draw[-{Stealth}] (m-1-1) -- (m-1-2);
\draw[-{Stealth}] (m-1-2) -- (m-2-2);
```

See



Adjust the matrix spacing before changing the arrows. The node names remain valid when rows or columns move.

12.8 Worked Project: A Three-by-Three Process

The matrix supplies alignment; local styles and arrows add meaning.

12.8.1 Stage 1: Define the Matrix

```
\matrix (m) [  
  matrix of nodes,  
  row sep=7mm,  
  column sep=8mm,  
  nodes={draw,minimum width=15mm,minimum height=8mm}  
]  
{  
  Input & Test & Output \\  
  A      & B      & C      \\  
  D      & E      & F      \\  
};
```

12.8.2 Stage 2: Highlight the Center

Replace the center entry with


```
A & |[fill=yellow]| B & C \\\
```

12.8.3 Stage 3: Add a Route

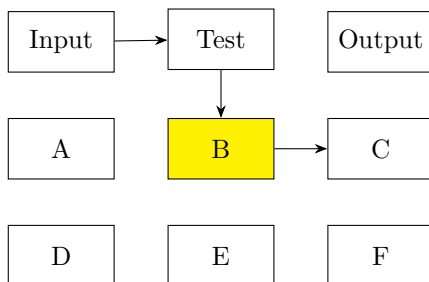
```
\draw[-{Stealth}] (m-1-1) -- (m-1-2);  
\draw[-{Stealth}] (m-1-2) -- (m-2-2);  
\draw[-{Stealth}] (m-2-2) -- (m-2-3);
```

12.8.4 The Complete Picture

```
\begin{tikzpicture}
  \matrix (m) [
    matrix of nodes,
    row sep=7mm,
    column sep=8mm,
    nodes={
      draw,
      minimum width=15mm,
      minimum height=8mm
    }
  ]
  {
    Input & Test & Output \\
    A & |[fill=yellow]| B & C \\
    D & E & F \\
  };

  \draw[-{Stealth}] (m-1-1) -- (m-1-2);
  \draw[-{Stealth}] (m-1-2) -- (m-2-2);
  \draw[-{Stealth}] (m-2-2) -- (m-2-3);
\end{tikzpicture}
```

See



Design

The matrix records alignment, the local fill records emphasis, and the arrows record a route. Each layer can be revised independently.

12.9 Debugging Workshop

12.9.1 Forgetting the Matrix Name

The reference `m-2-3` works only if the matrix was declared as `\matrix (m)`.

12.9.2 Forgetting a Row Terminator

A missing double backslash joins or corrupts rows. Check the end of every row, including the last one.

12.9.3 Unequal Cell Sizes

Set shared minimum dimensions before trying to repair individual cells.

12.9.4 Using Coordinates for Every Entry

If the structure is rectangular, write rows and columns in the matrix body and let TikZ calculate the positions.

12.10 Exercises

1. Create a two-by-three matrix of nodes.
2. Add uniform minimum width and height to every entry.
3. Change row spacing without editing the entries.
4. Highlight one corner cell and the center cell.
5. Name a matrix and connect its first and last entries.
6. Draw a three-by-three grid with touching borders.
7. Build a two-row flowchart using a matrix as the layout engine.
8. Add a second route through the worked project.

12.11 Solutions

12.11.1 Exercises 1–4

```
\matrix[matrix of nodes]
{
  A & B & C \\
  D & E & F \\
};
```

```
nodes={draw,minimum width=14mm,minimum height=8mm}
```

Exercise 3 changes only `row sep`.

```
|[fill=yellow]| A & B & C \\
D & |[fill=yellow]| E & F \\
```

12.11.2 Exercises 5–7

```
\matrix (m) [matrix of nodes]
{
  A & B & C \\
  D & E & F \\
};
\draw[-{Stealth}] (m-1-1) -- (m-2-3);
```

```
nodes={draw,minimum size=8mm},
row sep=-\pgflinewidth,
column sep=-\pgflinewidth
```

Exercise 7 uses the same technique as the connecting example: name the matrix, then connect entries by their **m-row-column** names.

For Exercise 8, add paths only. Do not move or rename any matrix entry.

12.12 ChatGPT Workshop

Ask ChatGPT to preserve the matrix structure while checking alignment, styles, and entry references.

1. Rewrite this coordinate-built grid as a named **matrix of nodes**.
2. Make every cell uniform by changing shared matrix options only.
3. Find every invalid **m-row-column** reference in this picture.
4. Highlight only the mathematically exceptional entries and remove decorative color from the others.
5. Change the row and column spacing without editing any cell or arrow.

12.13 Final Checklist

Before continuing to Chapter 13, check that you can

1. write rows and columns in a matrix body;
2. apply common options to every entry;
3. give cells uniform dimensions;
4. style one entry as an exception;
5. name a matrix and address an entry by row and column;
6. connect matrix entries with ordinary TikZ paths; and
7. revise spacing without moving individual cells.